Equations with Unknowns on Both Sides

Exam code: 0607

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Unknown on both sides

How do I solve linear equations with x terms on both sides?

- If a linear equation contains the unknown variable on **both** sides, **collect** the x terms together on **one side**
  - To do this, **choose** a side you wish to **remove** the x term from
    - this should be the side with the **lowest value** x term
  - Apply the **opposite** of that term to both sides
- For example, solve $4x - 7 = 11 + x$
- Let's remove the x term on the right-hand side
- x has been added, so subtract x from both sides

$$4x - 7 = 11 + x$$

$$(-x) \qquad\qquad (-x)$$

$$3x - 7 = 11$$

- There are no longer any x terms on the right
- The $4x$ has become $3x$ on the left
- This now has the same form as equations with **one variable**
  - Solve it by adding 7 then dividing by 3

$$3x - 7 = 11$$
$$3x = 18$$
$$x = 6$$

Help me understand

Examiner Tips and Tricks

In the exam, substitute your answer back into the original equation to check you got it right!

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$$4 - 5x = 6x - 29$$

Answer:

$5x$ is smaller than $6x$ so it's easier to remove this term first
Add $5x$ to both sides and simplify (collect like terms)

$$4 - 5x + 5x = 6x - 29 + 5x$$
$$4 = 11x - 29$$

This could also have been written as $4 = -29 + 11x$
Leave the x's on the right-hand side (avoid 'reflecting' the equation)
Get $11x$ on its own (by adding 29 to both sides)

$$4 + 29 = 11x$$

Work out $4 + 29$

$$33 = 11x$$

Get x on its own (by dividing both sides by 11)

$$\frac{33}{11} = x$$

Work out $33 \div 11$

$$3 = x$$

The answer currently has x on the right-hand side
At this point, you can reflect the equation to present your final answer as $x = \ldots$

$$x = 3$$

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